

Suppose a discrete uniform distribution from 0 to K . The pmf is $f(x; K) = \frac{1}{K+1}$, $x \in \{0, 1, \dots, K\}$. Suppose I have a sample of size 3 from the distribution: 276, 159, and 912.

1. Find the ML estimate of K . *Hint: The log-likelihood is discontinuous, so the usual optimization routine might mislead you. But the maximum is immediately apparent once you write out the likelihood. A verbal argument is okay, too!*
2. Find the method of moments estimate of K . *Hint: The mean of this uniform distribution is $\frac{K}{2}$. Set the sample mean equal to the model mean (i.e., $\frac{K}{2} = \text{avg}(x)$) and solve.*
3. Discuss any problems you notice with each estimator.